Primer on Frequency Domain Methods

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January 2021

Extracted from Appendix C of Confounding Dynamics, pages 53–60.

C Technical Preliminaries for Frequency Domain Methods

Elementary results concerning the theory of stationary stochastic processes and the residue calculus are necessary for grasping the z-transform approach (also known as frequency domain approach) advocated in the main text. The purpose of this appendix is to offer readers unfamiliar with the methods used to derive our results the minimal background necessary so that the paper is self-contained. This appendix introduces important theorems that are relatively well known but is by no means exhaustive. Interested readers are directed to Brown and Churchill (2013) and Whittle (1983) for good references on complex analysis and stochastic processes. Sargent (1987) provides a good introduction to these concepts and discusses economic applications.

C.1 Variance-Covariance Generating Function

Consider two-covariance stationary linear-Gaussian multivariate processes, $\{\omega_t, t \in \mathbb{Z}\}$ and $\{s_t, t \in \mathbb{Z}\}$, where the vector dimensions are $n \times 1$, and $m \times 1$, respectively. Let $\Upsilon_{\omega s}(j)$ denote the $m \times n$ unconditional covariance matrix between ω_t and s_{t-j}, for $j \in \mathbb{Z}$, formally

$$\Upsilon_{\omega s}(j) \equiv \mathbb{E}(\omega_t s_{t-j}^T) - \mathbb{E}(\omega_t)\mathbb{E}(s_{t-j}^T), \tag{C.1}$$

where T denotes transpose. The variance-covariance generating function is then defined as

$$g_{\omega s}(z) \equiv \sum_{j=-\infty}^{\infty} \Upsilon_{\omega s}(j) z^j, \tag{C.2}$$

where $g_{\omega s}(z)$ is an $m \times n$ matrix. When $\omega_t = s_t$ the function is referred to as the auto-covariance generating function and denoted by $g_{\omega\omega}(z)$, or $g_{ss}(z)$. An extensive treatment of the properties of the variance-covariance generating function can be found in Sargent (1987).

C.2 Wold Fundamental Representation Theorem

Much of the analysis in the main text is conducted in the space of lag polynomials without specific functional forms assumed (e.g., $ARMA(m, n)$). The Wold representation theorem allows for such a general specification.

Theorem C1. *[Wold Representation Theorem] Let $\{s_t\}$ be any $(n \times 1)$ covariance stationary stochastic process with $\mathbb{E}(s_t) = 0$. Then it can be uniquely represented in the form*

$$s_t = \eta_t + \Gamma^*(L)\hat{w}_t \tag{C.3}$$

where $\Gamma^*(L)$ is a matrix polynomial in the lag operator, and $\sum_{j=0}^{\infty} \Gamma_j^* \Gamma_j^{*T}$ is convergent. The process $\hat{w}_t$ is n-variate white noise with $\mathbb{E}(\hat{w}_t) = 0$, $\mathbb{E}(\hat{w}_t \hat{w}_t') = I_n$ and $\mathbb{E}(\hat{w}_t \hat{w}_{t-m}') = 0$ for $m \neq 0$. The process $\Gamma_0^* \hat{w}_t$ is the innovation in predicting s_t linearly from its own past:

$$\Gamma_0^* \hat{w}_t = s_t - \mathcal{P}(s_t | s_{t-1}, s_{t-2}, ...), \tag{C.4}$$

where $\mathcal{P}(\cdot)$ denotes linear projection. The process η_t is linearly deterministic; there exists an $n \times 1$ vector c_0 and $n \times n$ matrices C_s such that without error $\eta_t = c_0 + \sum_{s=1}^{\infty} C_s \eta_{t-s}$ and $\mathbb{E}[\hat{w}_t \eta_{t-m}'] = 0$ for all m.

The Wold representation theorem states that *any* covariance stationary process can be written as a linear combination of a (possibly infinite) moving average representation where the innovations are the linear forecast errors for s_t and a process that can be predicted arbitrarily well by a linear function of past values of s_t. The theorem is a *representation* determined by second moments of the stochastic process only and therefore may not fully capture the data generating process. For example, that the decomposition is linear suggests that a process could be deterministic in the strict sense and yet linearly non-deterministic; Whittle (1983) provides examples of such processes. The innovations in the Wold representation are generated by linear projections which need not be the same as the conditional expectation ($\mathbb{E}[s_t | s_{t-1}, s_{t-2}, ...]$). However, when working with linear Gaussian stochastic processes, as is standard in the rational expectations literature, the best conditional expectation coincides with linear projection.

As discussed in detail in the paper, the innovations derived from the Wold representation are an essential element of a rational expectations equilibrium in that they define the information set obtained from conditioning on current and past s_t's. Using the language of Rozanov (1967), the innovation sequence $\{\hat{w}_{t-j}\}_{j=0}^{\infty}$ of (C.4) are fundamental for the sequence $\{s_{t-j}\}_{j=0}^{\infty}$ if the Hilbert space generated by the observables is equivalent (in mean-square norm) to the Hilbert space generated by the innovations.

C.3 Canonical Factorization Let s_t be specified as

$$s_t = \Gamma(L)u_t, \tag{C.5}$$

where $\Gamma(L)$ is a lag polynomial matrix with square-summable coefficients, and u_t is a $m \times 1$ vector of i.i.d. Gaussian innovations with zero mean and variance-covariance matrix Σ_u. Testing for fundamentalness in a process like (C.5) can be done by checking for the invertibility of $\Gamma(z)$ for $|z| < 1$.[24] While testing for fundamentalness is straightforward, deriving the unique fundamental Wold representation is not. Here we rely on powerful factorization theorems. The following definition is taken from Whittle (1983). If $g(z)$ is a square matrix function of z, then the canonical factorization

$$g(z) = \Gamma^*(z)\Gamma^*(z^{-1})^T, \tag{C.6}$$

will refer to a factorization in which both $\Gamma^*(z)$ and its inverse have valid expansions in non-negative powers of z for $|z| \leq 1$; and $\Gamma^*(z^{-1})^T$ and its inverse have expansions in non-positive powers of z for $|z| \geq 1$. The canonical factorization of the variance-covariance generating function delivers the Wold fundamental representation of the time series. It yields both the fundamental moving average Wold representation, and the autoregressive representation that is consistent with span $\{s_{t-j}\}_{j=0}^{\infty}$ being equivalent to span $\{\hat{w}_{t-j}\}_{j=0}^{\infty}$.

There does not exist a general method to compute analytically the canonical factorization of any arbitrary matrix $g(z)$. Several methods that have been proposed work well when the knowledge of the structure of the matrix $g(z)$ is used. Rozanov (1967), page 47, proposes an algorithm to obtain $\Gamma^*(z)$ from $\Gamma(L)$ that can be used when $\Gamma(L)$ is known to have a finite number of isolated singularities $\lambda_1, \lambda_2, ..., \lambda_p$ inside the unit circle. In the case of $n = 2$, and one singularity at $|\lambda| < 1$, one has that

$$\Gamma^*(z) = \Gamma(z)W_\lambda B_\lambda(z), \tag{C.7}$$

where $B_\lambda(z)$ is the Blaschke matrix

$$B_\lambda(z) = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1-z\lambda}{z-\lambda} \end{bmatrix}. \tag{C.8}$$

The constant matrix W_λ is a unitary matrix, so that $W_\lambda^{-1} = W_\lambda^T$, whose columns are the left singular vectors of the system matrix $\Gamma(z)$ evaluated at λ. While the inverse of $\Gamma(\lambda)$ does not exist, the pseudo-inverse (also known as the Moore-Penrose inverse) always exists and it is given by

$$\Gamma(\lambda)^+ = U\Sigma V^T, \tag{C.9}$$

where $V\Sigma U^T$ is the singular value decomposition of the matrix $\Gamma(\lambda)$. The left singular vectors of $\Gamma(\lambda)$ are the columns of U. The following result, which can be showed by using widely available formulas for the singular value decomposition of 2 by 2 matrices is used in the proof of Lemma A1.

Lemma C1. *Consider the matrix*

$$N = \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix}. \tag{C.10}$$

Then the matrix U in the singular value decomposition of $N = V\Sigma U^T$, is

$$U = \frac{1}{\sqrt{c^2+d^2}} \begin{bmatrix} \frac{cd}{|d|} & -\frac{cd}{|c|} \\ \frac{d^2}{|d|} & \frac{c^2}{|c|} \end{bmatrix}. \tag{C.11}$$

[24] There are many ways to test for fundamentalness, see Fernandez-Villaverde, Rubio, Sargent, and Watson (2007).

C.4 Riesz-Fischer Theorem

Theorem C2. *[Riesz-Fischer] Let $D(\sqrt{r})$ denote a disk in the complex plane of radius $\sqrt{r}$ centered at the origin. There is an equivalence (i.e. an isometric isomorphism) between the space of r-summable sequences $\sum_j r^j |f_j|^2 < \infty$ and the Hardy space of analytic functions $f(z)$ in $D(\sqrt{r})$ satisfying the restriction*

$$\frac{1}{2\pi i} \oint f(z) f(r z^{-1}) \frac{dz}{z} < \infty$$

where $\oint$ denotes (counterclockwise) contour integration around $D(\sqrt{r})$. An analytic function satisfying the above condition is said to be r-integrable.[25]

The Riesz-Fischer theorem ensures that one can work either in the space of infinite sequence of square-summable matrix coefficients $\{..., Q_{-1}, Q_0, Q_1, ...\}$, or in the space of complex-valued analytic functions $Q(z) = \sum_{j=-\infty}^{\infty} Q_j z^j$, since the two spaces are equivalent.

C.5 Wiener-Kolmogorov Prediction Formula

Consider the problem of computing the linear least-squares estimate for ω_{t+j}, denoted by $\hat{\omega}_{t+j}$, conditional on the realized history $\{s_{t+j}\}_{j=0}^{\infty}$. The solution to the problem consists in the sequence of real valued matrices $\{\Pi_j\}_{j=0}^{\infty}$, or, equivalently, the complex-valued function $\Pi(z)$, such that

$$\hat{\omega}_{t+j} = \sum_{j=0}^{\infty} \Pi_j s_{t-j}. \tag{C.12}$$

We assume that all the processes have zero unconditional mean. The following result is a version of the Wiener-Kolmogorov prediction formula taken from Whittle (1983).

Theorem C3. *Suppose that $g_{ss}(z)$ has the canonical factorization*

$$g_{ss}(z) = \Gamma^*(z) \Gamma^*(z^{-1})^T. \tag{C.13}$$

then the generating function $\Pi_j(z) = \sum_{i=0}^{\infty} \Pi_i z^i$ of the optimal estimate $\hat{\omega}_{t+j}$ is

$$\Pi_j(z) = \left[z^{-j} g_{\omega s}(z) \left(\Gamma^*(z^{-1})^T \right)^{-1} \right]_+ \Gamma^*(z)^{-1}. \tag{C.14}$$

Proof. The minimization of the the means squared forecast errors $\mathbb{E}(\hat{\omega}_t - \omega_t)^2$ leads to the set of first order conditions

$$\sum_{k=0}^{\infty} \Upsilon_{ss}(i-k) = \Upsilon_{\omega s}(i-j), \quad \text{for} \quad i = 0, 1, 2, ... \tag{C.15}$$

Multiplying both sides by z^i and adding over all integral i's one gets the Wiener-Hopf relationship

$$\Pi_j(z) g_{ss}(z) = z^{-j} g_{\omega s}(z) + h(z), \tag{C.16}$$

where $h(z) = \sum_{i=-\infty}^{-1} h_i z^i$, is an unknown matrix series in negative powers of z. If we post-multiply both sides by $\left[\Gamma^*(z^{-1})^T \right]^{-1}$, we obtain

$$\Pi_j(z) \Gamma^*(z) = z^{-j} g_{\omega s}(z) \left[\Gamma^*(z^{-1})^T \right]^{-1} + h(z) \left[\Gamma^*(z^{-1})^T \right]^{-1}, \tag{C.17}$$

Note that both sides can be represented as matrix series in powers of z whose coefficients have to obey the above relationship. By construction, the left hand side has only positive powers of z, while the second term of the right hand side has only negative powers of z. If we apply the annihilating operator on both sides we thus get

$$\Pi_j(z) \Gamma^*(z) = \left[z^{-j} g_{\omega s}(z) \left[\Gamma^*(z^{-1})^T \right]^{-1} \right]_+. \tag{C.18}$$

[25] This theorem is usually proved for the case $r = 1$ and for functions defined on the boundary of a disk. For further exposition see Sargent (1987).

The last step consists in post-multiplying both sides by $\Gamma^*(z)^{-1}$ so that (C.14) obtains. □

The requirement that $\Gamma^*(z)$ should be the canonical factorization of $g_{ss}(z)$ is essential in two steps. First, from it we ensure that $\left[\Gamma^*(z^{-1})^T\right]^{-1}$ has an expansion in non-positive powers of z, which means that the term $h(z)\left[\Gamma^*(z^{-1})^T\right]^{-1}$ disappears when the annihilating operator is applied. Second, we also ensure that $\Gamma^*(z)^{-1}$ has an expansion in non-negative powers of z, which result in $\Pi(z)$ having an expansion in non-negative powers of z only, as required.

C.6 Whiteman (1983) Solution Method: Existence and Uniqueness of REE We use the existence and uniqueness criteria of Whiteman (1983) developed for linear, rational expectations equilibria. The following works through the three relevant cases–existence but no uniqueness, no existence, and existence-uniqueness. Consider the following generic rational expectations model

$$\zeta\tilde{\zeta}\mathbb{E}_t y_{t+1} - (\zeta + \tilde{\zeta})y_t + y_{t-1} = \theta_t, \qquad \theta_t = A(L)\varepsilon_t, \qquad \varepsilon_t \overset{iid}{\sim} N(0,1) \tag{C.19}$$

where ε_t is assumed to be fundamental for θ_t (i.e., $A(L)$ is assumed to have a one-sided inverse in non-negative powers of L). Following the solution principle, we will look for a solution that is square-summable in the Hilbert space generated by the fundamental shock ε, $y_t = Q(L)\varepsilon_t$ (third tenet). If we invoke the optimal prediction formula (C.14), then $\mathbb{E}_t y_{t+1} = [Q(L)/L]_+\varepsilon_t = L^{-1}[Q(L) - Q_0]\varepsilon_t$. Together with the fourth tenet of the solution principle (i.e., that the rational expectation restrictions hold for all realizations of ε), this implies that (C.19) can be written in z-transform as

$$z^{-1}[Q(z) - Q_0]\zeta\tilde{\zeta} - (\zeta + \tilde{\zeta})Q(z) + zQ(z) = A(z)$$

Multiplying by z and rearranging delivers

$$Q(z) = \frac{zA(z) + Q_0}{(\zeta - z)(\tilde{\zeta} - z)} \tag{C.20}$$

Appealing to the Riesz-Fischer Thereom, square-summability (stationarity) in the time domain is tantamount to analyticity of $Q(z)$ on the unit disk. The function $Q(z)$ is analytic at z_0 if it is continuously (complex) differentiable in an open neighborhood of z_0.[26] Any rational function $(f(z)/g(z))$ where $f(\cdot)$ and $g(\cdot)$ are polynomials will be analytic on the unit disk provided $g(z) \neq 0$ at any point inside the unit circle. The extent to which this is true for $Q(z)$ depends upon the parameters ζ and $\tilde{\zeta}$.

As shown in Whiteman (1983), there are three cases one must consider. First, assume that $|\zeta| > 1$ and $|\tilde{\zeta}| > 1$. Then (C.20) is an analytic function on $|z| < 1$ and the representation is given by

$$y_t = \left[\frac{LA(L) + Q_0}{(\zeta - L)(\tilde{\zeta} - L)}\right]\varepsilon_t \tag{C.21}$$

For any finite value of Q_0, this is a solution that lies in the Hilbert space generated by $\{\theta_t\}$ and satisfies the tenets of the solution principle. Thus, we have existence but not uniqueness because Q_0 can be set arbitrarily.

The second case to consider is $|\tilde{\zeta}| < 1 < |\zeta|$. In this case, the function $Q(z)$ has an isolated singularity at ζ, implying that $Q(z)$ is not analytic on the unit disk. In this case, the free parameter Q_0 can be set to remove the singularity at ζ by setting Q_0 in such a way as to cause the residue of $Q(\cdot)$ to be zero at ζ

$$\lim_{z\to\zeta}(\tilde{\zeta} - z)Q(z) = \frac{\zeta A(\zeta) + Q_0}{\zeta - \tilde{\zeta}} = 0$$

Solving for Q_0 delivers $Q_0 = -\zeta A(\zeta)$. Substituting this into (C.21) yields the following rational expectations equi-

[26] Analytic is synonymous with holomorphic, regular and regular analytic.

librium

$$y_t = \left[\frac{LA(L) - \zeta A(\zeta)}{(\zeta - L)(\tilde{\zeta} - L)}\right]\varepsilon_t. \tag{C.22}$$

The function $Q(z)$ is now analytic and (C.22) is the unique solution that lies in the Hilbert space generated by $\{\theta_t\}$. The solution is the ubiquitous Hansen-Sargent prediction formula that clearly captures the cross-equation restrictions that are the "hallmark of rational expectations models" [Hansen and Sargent (1980)].[27]

The final case to consider is $|\zeta| < 1$ and $|\tilde{\zeta}| < 1$. In this case, (C.20) has two isolated singularities at ζ and $\tilde{\zeta}$, and Q_0 cannot be set to remove both singularities.[28] Hence in this case, there is no solution in the Hilbert space generated by $\{\theta_t\}$ and we do not have existence.

C.7 Annihilating Operator Let $H(z) = \sum_{m=-\infty}^{+\infty} H_m z^m$. The following Lemma is due to Hansen and Sargent (1980).

Lemma C2. *Let $H(z)$ be a regular function in $|z| < 1$, with at most a finite number of singularities $z_1, z_2, ..., z_k$ in $|z| < 1$. Let $\pi_1(z)$, $\pi_2(z)$,..., $\pi_k(z)$ denote the principal parts of the Laurent series expansion of $H(z)$ around the singularities. Then*

$$[H(z)]_+ = H(z) - \sum_{j=1}^{k} \pi_j(z). \tag{C.23}$$

The Laurent series expansion of $H(z)$ around the singularity z_j is

$$H(z) = \sum_{m=-\infty}^{+\infty} H_m (z - z_j)^m, \tag{C.24}$$

and its principal part is given by

$$\pi_j(z) = \sum_{-\infty}^{-1} H_m (z - z_j)^m. \tag{C.25}$$

To compute the principal part at z_j, first compute the constant $\hat{\pi}_j$ as

$$\hat{\pi}_j = \lim_{z \to z_j} (z - z_j) H(z), \tag{C.26}$$

and then set

$$\pi_j(z) = \hat{\pi}_j (z - z_j). \tag{C.27}$$

In this section, we first establish notation and introduce relevant mathematical definitions. We then formalize the notion of confounding dynamics, presenting a simple example that shows the mechanism at work.

C.8 Frequency Domain and Confounding Dynamics We conclude this primer with a general discussion of how the results presented above are useful in the analysis of equilibria with incomplete information, and how they greatly facilitate the study of equilibria with confounding dynamics. Throughout the paper, we work in the space of polynomials in the lag operator L with square-summable coefficients that operate on Gaussian random variables. In our framework, any stochastic process ω_t can always be written as

$$\omega_t = Q(L)\varepsilon_t = \sum_{j=0}^{\infty} Q_j L^j \varepsilon_t, \tag{C.28}$$

where $\sum_{j=0}^{\infty} |Q_j|^2 < \infty$, and $\varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon)$, are innovations identically and independently distributed over time. In linear-Gaussian environments, working with representations of the form of (C.28), and their functional equivalents,

[27] Our methodology can also handle unit roots. For example, suppose $\theta_t = (1 - L)A(L)\varepsilon_t$. The solution, $Q(L)\varepsilon_t$, would then inherit the unit root via the cross-equation restriction.

[28] As discussed by Whiteman (1983), the problem remains even if $\zeta = \tilde{\zeta}$.

has three advantages for analyzing rational expectations models with incomplete information.[29]

First, representation (C.28) is general in the sense that it can accommodate both autoregressive (AR) and moving-average (MA) components of any order. This is especially useful when searching for an equilibrium because it avoids the need to specify a conjecture with a specific ARMA order. Regardless of the complexity of the equilibrium conditions that emerge in models of dispersed information (e.g., infinite regress in expectations), the solution will take the form of (C.28).[30]

Second, the Wold Representation Theorem ensures that processes like ω_t can always be written *uniquely* as a linear combination of a moving average representation where the innovations are the linear forecast errors for ω_t, conditional on any linear-Gaussian information set [Brockwell and Davis (1987)]. That is, the Wold representation establishes the invertibility of $Q(L)$ and one may write $Q(L)^{-1}\omega_t = \varepsilon_t$, which implies that the space spanned by $\{\omega_t, \omega_{t-1}, ...\}$ is equivalent (in mean-square norm) to the space spanned by $\{\varepsilon_t, \varepsilon_{t-1}, ...\}$. Consequently, one can apply the optimal prediction formulas derived by Wiener-Kolmogorov [Whittle (1983)] to compute the conditional expectation of processes like ω_t.

Third, the Riesz-Fischer Theorem [see Sargent (1987)] establishes an isometric, isomorphic mapping from the space of lag polynomials with square-summable coefficients $Q(L)$ to the space of analytic complex-valued functions, where (C.28) is represented as $Q(z)$, but with $z \in \mathbb{C}$. In several key steps of the analysis in this paper we find it convenient to exploit the properties of such functions, which allows us to derive simple existence and uniqueness conditions for rational expectations equilibria with incomplete and dispersed information following Whiteman (1983). In a slight abuse of notation, we employ L and z interchangeably when working in the space of analytic functions.

We now restrict our focus to the formulation of optimal prediction formulas, which is where confounding dynamics emerge.

Suppose that we would like to formulate the prediction of ω_{t+j} so as to minimize the mean-squared forecast error, conditional on the observation of the history of a $n \times 1$ vector of variables, s_t, up to time t. To denote such history, we use the compact notation, $s^t \equiv \{s_{t-j}\}_{j=0}^{\infty}$. Let

$$s_t = \Gamma(L)u_t, \tag{C.29}$$

where $\Gamma(L)$ is an $n \times m$ matrix with each element being a square-summable lag polynomial in non-negative powers of L, and u_t an $m \times 1$ vector of i.i.d. Gaussian shocks with variance-covariance normalized to the $m \times m$ identity matrix. Let $g_{ss}(z)$ be the variance-covariance generating function for the process s_t,[31] then one has that $g_{ss}(z) = \Gamma(z)\Gamma(z^{-1})^{\top}$. Similarly, one can define the covariance generating function between the joint processes ω_t and s_t, which is given by $g_{\omega s}(z) = Q(z)\sigma_{\varepsilon}\Gamma(z^{-1})^{\top}$. The prediction for ω_{t+j} that minimizes the mean squared forecast error corresponds to a linear combination of current and past realizations of s_t, denoted by

$$\mathcal{P}(\omega_{t+j}|s^t) = \Pi(L)s_t. \tag{C.30}$$

Here, $\Pi(L)$ is a $1 \times n$ vector of square-summable lag polynomials in non-negative powers of L, whose form is provided by the Wiener-Kolmogorov formula

$$\Pi(L) = \left[L^{-j} g_{\omega s}(L)\left(\Gamma^*(L^{-1})^{\top}\right)^{-1}\right]_{+} \Gamma^*(L)^{-1}. \tag{C.31}$$

Expression (C.31) has several moving parts that require some unpacking. Let us first consider the special case $s_t = \omega_t$, which implies $\Gamma(L) = Q(L)$ and $u_t = \varepsilon_t$, so that the prediction problem is one in which we would like to predict the future realizations ω_{t+j}, for $j \geq 1$, using its own past, ω^t. Given the form of ω_t in (C.28), the best prediction is one

[29] We are not the first to highlight these advantages [see, Hansen and Sargent (1980), Futia (1981), Townsend (1983b), Kasa (2000)].

[30] Townsend (1983a) elaborates extensively on the advantages of using representations such as (C.28) when solving for equilibria that harbor an infinite regress in expectations. In Section 3.5 we show the complex form that the infinite regress in expectations takes in our framework.

[31] Recall from above that the variance-covariance generating function of a stationary Gaussian process is defined as the Fourier transform of its correlogram, which is the collection of covariances at all horizons.

that carries $Q(L)$, j periods forward and uses the best estimates of the infinite history of innovations $\{\varepsilon_{t+j}, \varepsilon_{t+j-1}, ...\}$, to compute ω_{t+j}. The best estimates of $\{\varepsilon_{t+j}, \varepsilon_{t+j-1}, ..., \varepsilon_{t+1}\}$ are clearly equal to the unconditional average, 0. It follows that they should not appear in the optimal prediction. This is achieved by the operator $[\ ]_+$ in (C.31), known as the "annihilating operator", which instructs us to ignore the first j coefficients of its argument.

The best estimates for $\{\varepsilon_t, \varepsilon_{t-1}, \varepsilon_{t-2}, ...\}$, on the other hand, are usually not zero, and correspond to the innovations in the Wold fundamental representation for ω_t. The decomposition of the information set into its Wold fundamental representation is achieved by $\Gamma^*(L)$ in (C.31). Whether $\Gamma^*(L)$ is equal to $\Gamma(L)$ depends on the invertibility properties of the analytic function $\Gamma(z)$. For our special case, if $Q(z)$ is invertible for z inside the unit circle, then the Wold fundamental representation corresponds exactly with the history $\{\varepsilon_t, \varepsilon_{t-1}, \varepsilon_{t-2}, ...\}$, and the prediction formula $\mathcal{P}(\omega_{t+j}|\omega^t) = \tilde{\Pi}(L)\omega_t$, has

$$\tilde{\Pi}(L) = [L^{-j}Q(L)]_+ Q(L)^{-1}. \tag{C.32}$$

Note that all the steps just described are clearly at work here: L^{-j} carries the function $Q(L)$ j periods forward; the operator $[\ ]_+$ sets the estimates of innovations from $t+1$ to $t+j$ to zero, by annihilating Q_0, Q_1,...,Q_{j-1}; finally, $Q(L)^{-1}$ makes sure that once $\Pi(L)$ is multiplied by ω_t, the Wold fundamental innovations $\{\varepsilon_t, \varepsilon_{t-1}, \varepsilon_{t-2}, ...\}$ result.

Expression (C.32) is a special case of (C.31) because, in general, $\Gamma(z)$ is not invertible for z inside the unit circle, and solving for $\Gamma^*(z)$ is at the core of the solution to the prediction problem. Formally, $\Gamma^*(z)$ corresponds to the "canonical" factorization of the variance-covariance generating function $g_{ss}(z)$, so that

$$g_{ss}(z) = \Gamma^*(z)\Gamma^*(z^{-1})^\top. \tag{C.33}$$

The canonical factorization answers the question: What space is spanned by the observables s^t? If $\Gamma(z)$ is invertible in non-negative powers of z, then s^t will span the innovations u^t. If $\Gamma(z)$ is non-invertible, then $\Gamma^*(z)$ must be determined and it will span a space that is strictly smaller than that spanned by u^t. The existence of $\Gamma^*(z)$ is guaranteed by the Wold (fundamental) Representation Theorem. The computation of $\Gamma^*(z)$ can be quite involved, however, and there exist several methods for achieving it. In our setting, we follow the steps outlined in Rozanov (1967).[32]

The most straightforward reason for $\Gamma(z)$ to be non-invertible for prediction purposes, is when $m > n$. In such case, the dimension of the vector of underlying shocks, u_t, is greater than the vector of signals, s_t, at any t. This source of non-invertibility is what is typically assumed in the incomplete information literature. A different, and often more subtle reason, is when the elements in $\Gamma(z)$ combine in a way to create an "internal" source of non-invertibility. While the analysis of Section 2 formally illustrates this claim, the easiest way to see this is to consider the case of $m = n$, so that $\Gamma(z)$ is a square matrix. $\Gamma(z)$ is non-invertible for prediction purposes if its determinant vanishes at one or more points inside the unit circle. In this case, despite having the same number of shocks and signals, non-invertibility stems from the way signals combine over time and become themselves a source of noise.

We are interested in this internal source of non-invertibility, which we term "Confounding Dynamics". Confounding dynamics naturally arise also when $m > n$, in which case they compound with the first source of non-invertibility, but their characterization can be substantially more involved. Because our focus is on showing that confounding

[32] Whittle (1983) shows that the computation of $\Gamma^*(L)$ is the dual to the solution of the nth order Ricatti equation in the Kalman filter approach to optimal linear-quadratic prediction. In both cases the objective is to figure out the variance-covariance of the optimal prediction errors. Hence, *if* the Ricatti equation is solved, then $\Gamma^*(z)$ can be determined; and *if* $\Gamma^*(L)$ is computed, then a solution to the Ricatti equation can be determined. In this paper we take the latter approach, while other authors, most notably Huo and Takayama (2016), take the former. Whether it is easier to solve the Ricatti equation or to obtain the canonical factorization often depends on the application at hand, and whether one is looking for an analytical or a numerical solution. The canonical factorization, however, has a wider scope of applicability. Whittle (1983) presents a generalized approach – which he terms the Hamiltonian approach – where the prediction problem is solved by the canonical factorization of a matrix function obtained by augmenting the plant and observation equations of a Kalman system with the saddle point conditions of an Hamiltonian optimization problem. Huertgen, Hoffmann, Rondina, and Walker (2016) make an explicit connection between these solution methodologies.

dynamics can endogenously emerge in equilibrium, in the paper we focus mostly on the $m = n$ case. In Appendix B.8 we analyze an example where the two sources of non-invertibility are simultaneously present, while we provide a more formal treatment of the $m > n$ case in Appendix B.3. The following definition formalizes the above discussion.